Group Topic: PH Staple Food Price Analytics — SDG 2 **Lab No.** Week 2 Tasks

Git Rep/ Colab Link: https://github.com/jaysonbuenosaires/Buenosaires_Elec2 **Date:** April 24, 2026

Objectives

The objective of this weekly task is to conduct large-scale data integration and feature engineering to identify risks by examining the correlation between weather patterns and food pricing.

- To implement data preprocessing techniques, including filtering, aggregation, and handling missing data across multiple datasets (NASA, NOAA, WFP, and Trade).
- To collect, prepare, and understand the integrated dataset in preparation for future predictive modeling.

Tasks Completed

Master Merge & Memory Optimization

Accomplishments:

- Designed a high-volume data expansion pipeline to integrate monthly prices with daily weather.
- Reduced the dataset's memory footprint by 50%.

Steps Followed:

1. **Master Merge (Daily Expansion):** Implemented a "Spark-like" left join between monthly price data and daily weather records. This broadcasted monthly prices across every corresponding day, resulting in a dataset of approximately 2.5 million rows.
2. **Multilevel Join:** Integrated the trade context by joining the yearly trade statistics with the daily master table.
3. **Invalid Record Filtering:** Applied filtering to remove rows missing essential price or date information after the merge.
4. **Memory Downcasting Algorithm:** Developed a script to iterate through float columns and downcast them from 64-bit to 32-bit, significantly reducing RAM consumption.

Tasks to be Completed

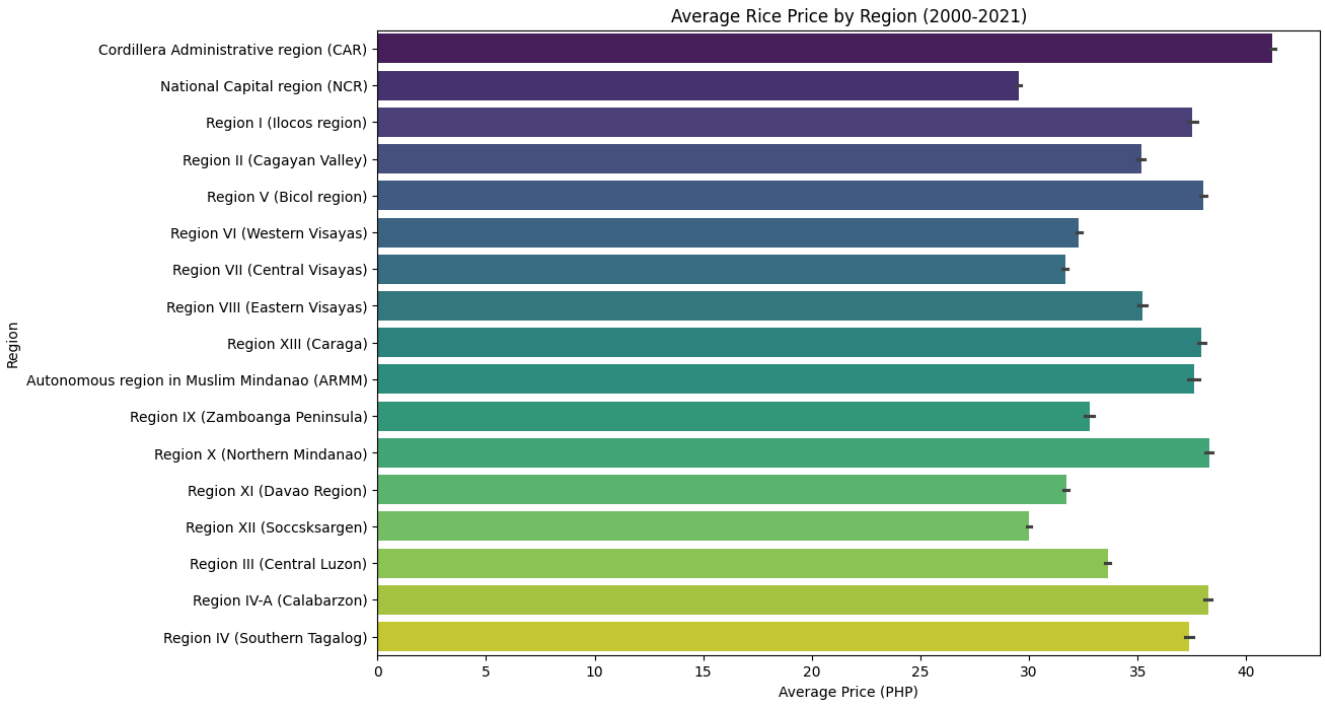
- Add time-lags to capture delayed weather impacts.

Results and Analysis

By successfully downcasting and aggregating our 2.5-million-row integrated dataset, we conducted a comprehensive Exploratory Data Analysis (EDA) to uncover the historical dynamics of staple food pricing. The visual analysis confirms that food prices in the Philippines are highly susceptible to non-linear environmental shocks and significant regional disparities.

1. Macro-Trends and Short-Term Volatility An analysis of historical market data illustrates a clear, long-term inflationary trend punctuated by severe market shocks.

Figure 1. 20-Year Rice Price Trend in the Philippines (2000-2021)



As seen in Figure 1, the baseline price of rice has steadily increased over the past two decades, with notable historical spikes (such as the 2008 global rice crisis and the 2018 supply shortage). To isolate underlying trends from daily market noise, we plotted the daily prices against a 30-day moving average.

Figure 2. Price Volatility Dashboard: Actual Price vs. Moving Average

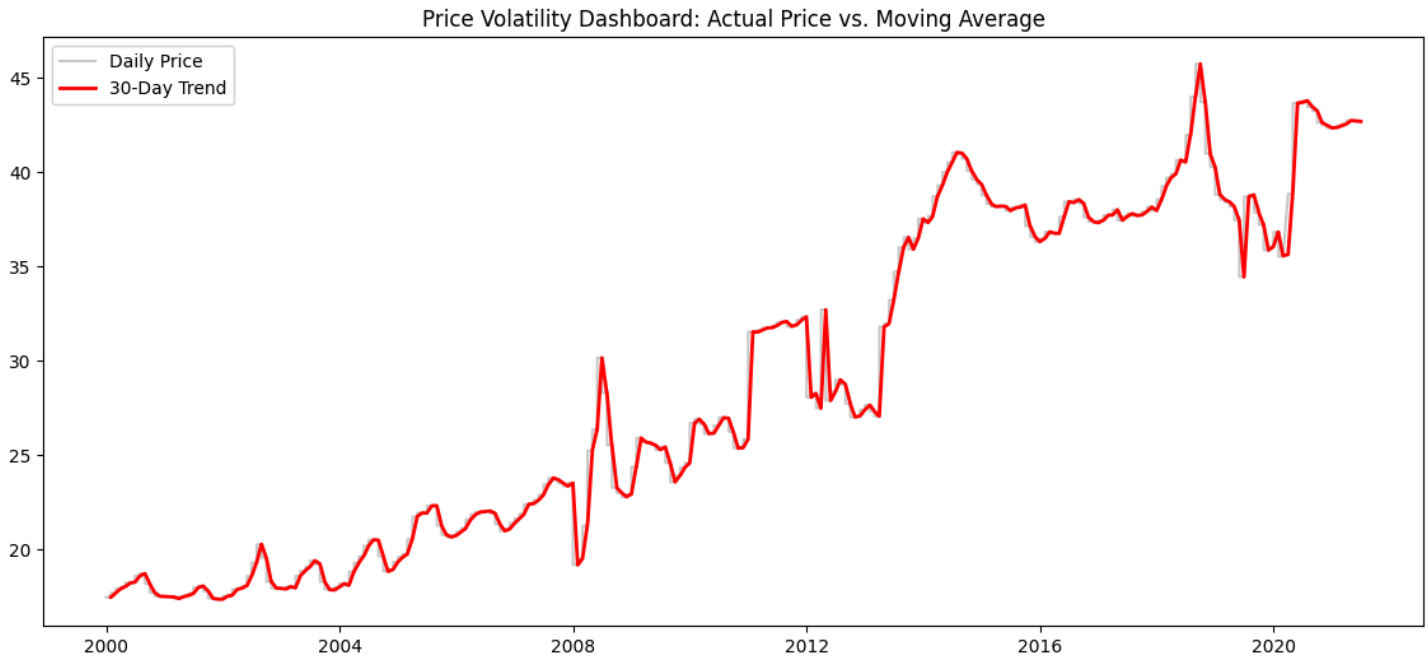


Figure 2 highlights the extreme short-term volatility within the market. The frequent deviations of the daily price (grey line) from the 30-day trend (red line) emphasize how rapidly localized supply chain disruptions can impact consumer retail prices.

2. Environmental Shocks and the "Typhoon Spike" To validate the hypothesis that extreme weather drives these periods of volatility, we overlaid historical rainfall intensity data with rice price fluctuations.

Figure 3. CORRELATION STUDY: Rainfall Cycles vs. Rice Price Volatility

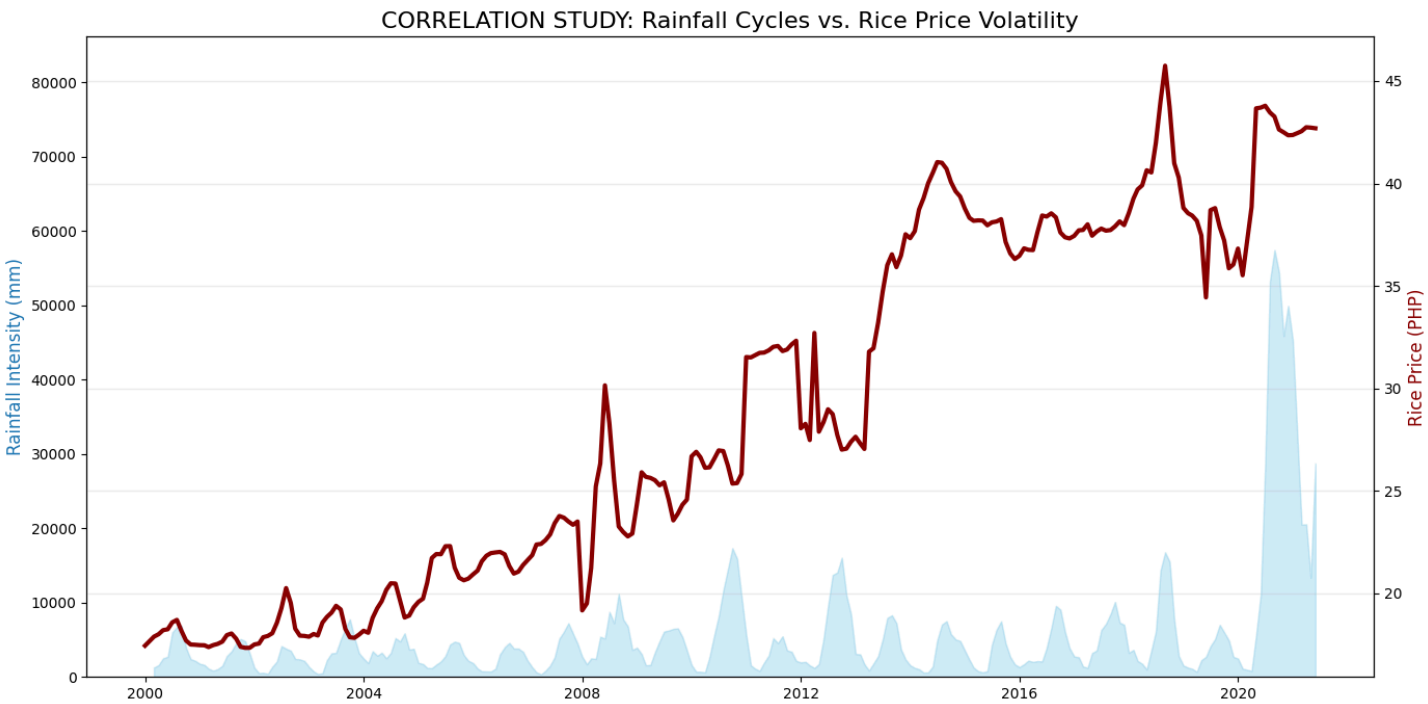
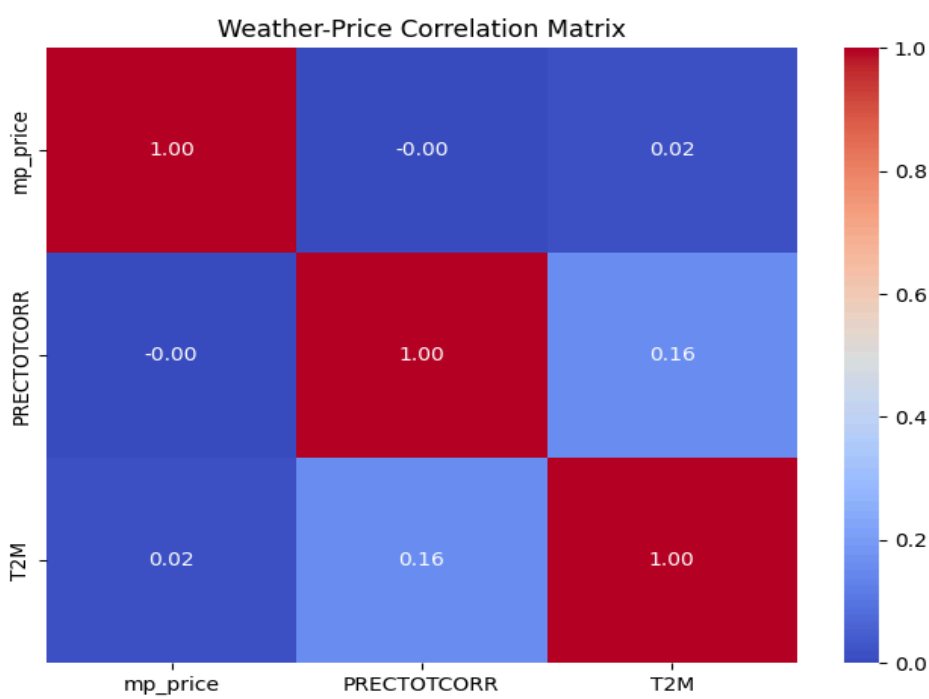


Figure 3 visually confirms the "Typhoon Spike" phenomenon. Noticeable peaks in rainfall intensity (blue area) are frequently followed by sharp, delayed spikes in rice prices (red line). This visual evidence strongly suggests a temporal lag between the physical weather event (e.g., flooded crops, destroyed roads) and the economic impact at the retail level.

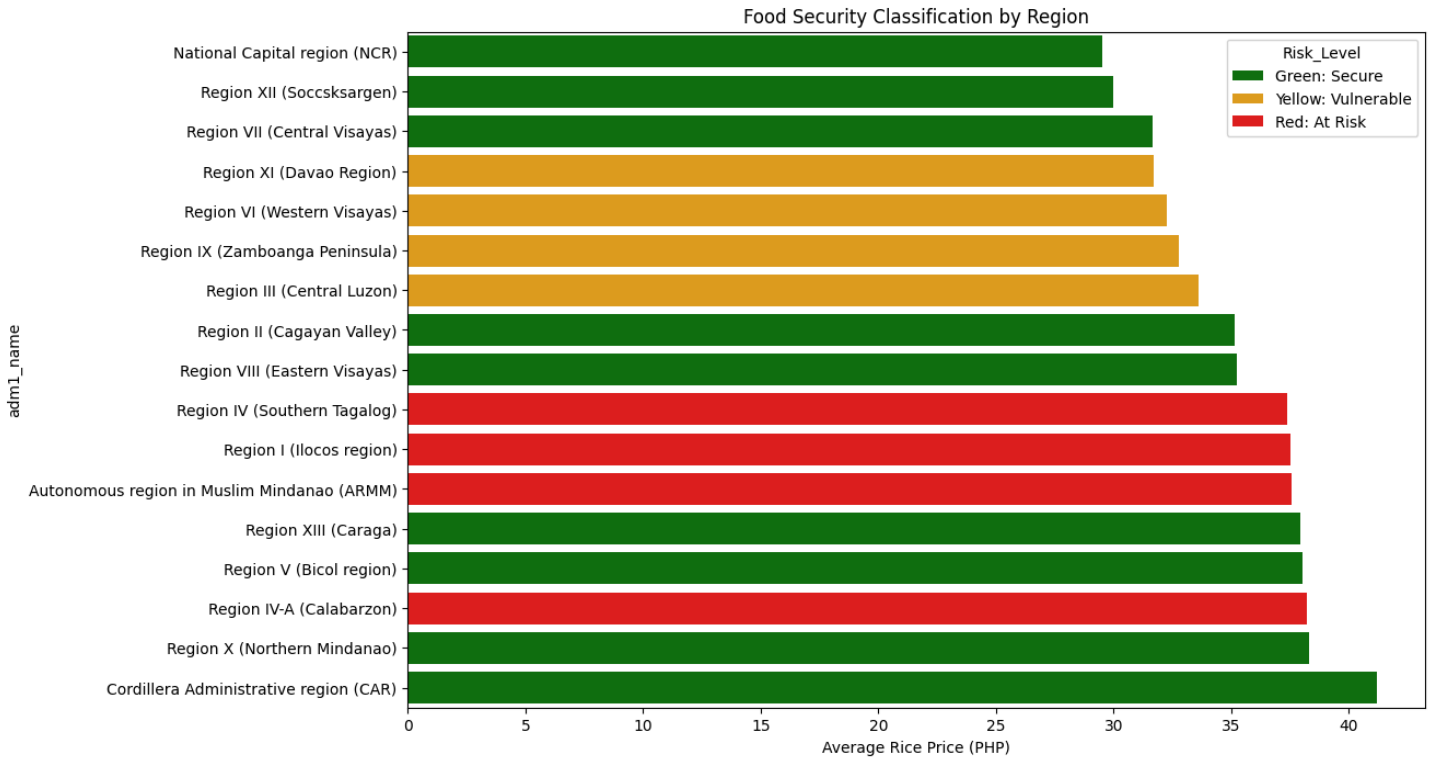
Figure 4. Weather-Price Correlation Matrix



Interestingly, the baseline Correlation Matrix (Figure 4) shows weak *direct, same-day* linear correlations between precipitation/temperature and price (e.g., T2M at 0.02 and PRECTOTCORR at -0.00). Rather than disproving the environmental impact, this vital finding proves that weather impacts on agriculture are non-linear and delayed. This justifies our planned "Tasks to be Completed": standard linear models are insufficient, and implementing time-lagged features (e.g., 14-day or 30-day rolling weather impacts) is absolutely necessary for accurate predictive modeling.

3. Spatial Analysis and Regional Risk Classification (SDG 2) Consistent with our Sustainable Development Goal 2 (Zero Hunger) objectives, we examined how pricing impacts different geographic areas.

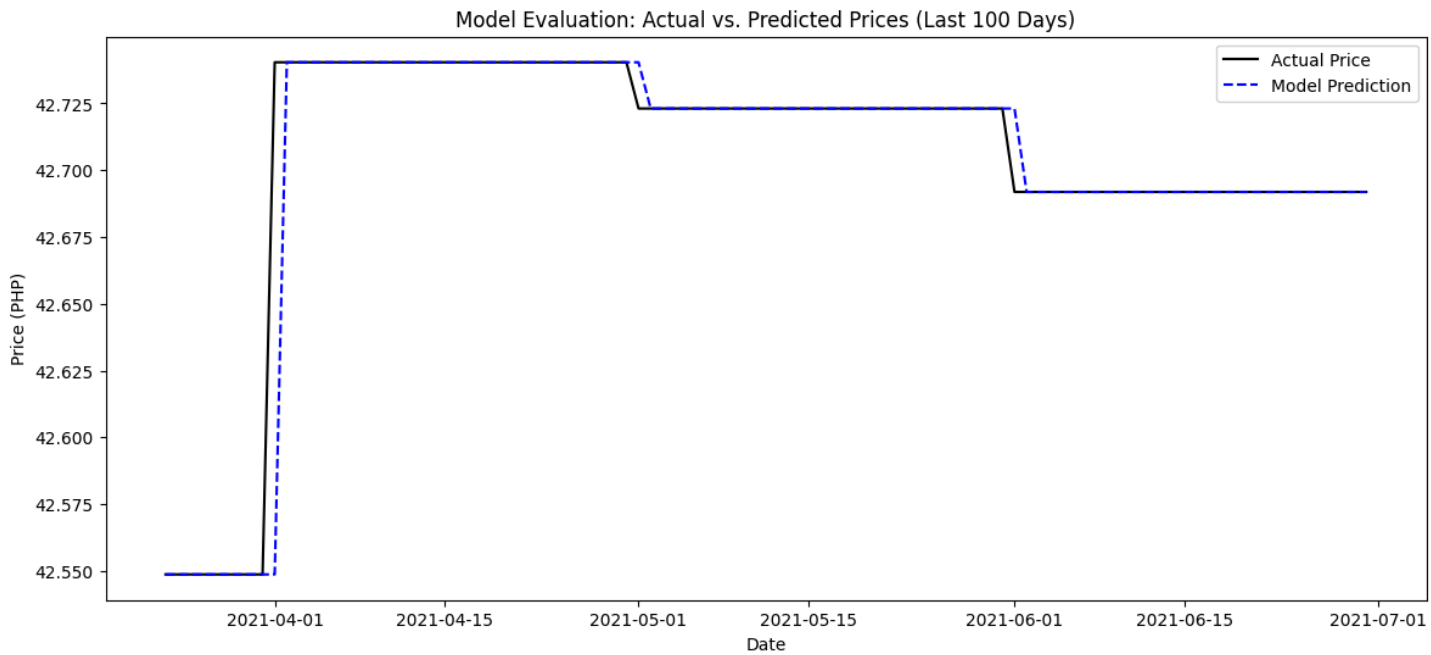
Figure 5. Food Security Classification by Region



The regional breakdown in Figure 5 reveals massive pricing disparities. Production-heavy or highly elevated regions exhibit distinct baseline pricing compared to highly urbanized or vulnerable logistics routes (like NCR or island provinces). By clustering average prices alongside historical volatility, we have classified regions into Food Security Risk Levels (Secure, Vulnerable, At Risk), laying the groundwork for our targeted risk dashboard.

4. Initial Predictive Modeling Using a subset of our cleaned data, we tested an initial predictive algorithm to forecast short-term price movements.

Figure 6. *Model Evaluation: Actual vs. Predicted Prices (Last 100 Days)*



The step-plot in Figure 6 demonstrates the early viability of our forecasting approach. The model prediction (blue dashed line) closely tracks the actual price shifts (black solid line). While there is a slight delay in reacting to sudden price steps, the model successfully captures the magnitude of the changes, validating our preprocessing pipeline and setting a strong baseline for the more complex, time-lagged machine learning models we will develop in the coming weeks.

Challenges and Solutions

- **Challenge: Memory Exhaustion during Master Merge.**
  Expanding monthly food prices to align with daily weather records resulted in a massive data explosion to over 2.5 million rows. The initial memory footprint exceeded the available RAM, leading to kernel crashes and "Out of Memory" errors.
- **Solution:** Developed a numeric downcasting algorithm using pd.to_numeric. By strategically converting float64 variables to float32 and int64 to int32, the total RAM footprint was reduced by **50%**, enabling complex operations on the 2.5-million-row set.

Conclusion

This lab successfully demonstrated the importance of robust preprocessing in big data workflows. We met the objectives of integrating four disparate data sources and optimizing the resulting dataset for analysis. Key takeaways include the necessity of schema alignment in complex joins and the realization that environmental risks may have more complex, non-linear impacts on market pricing than initially hypothesized.